

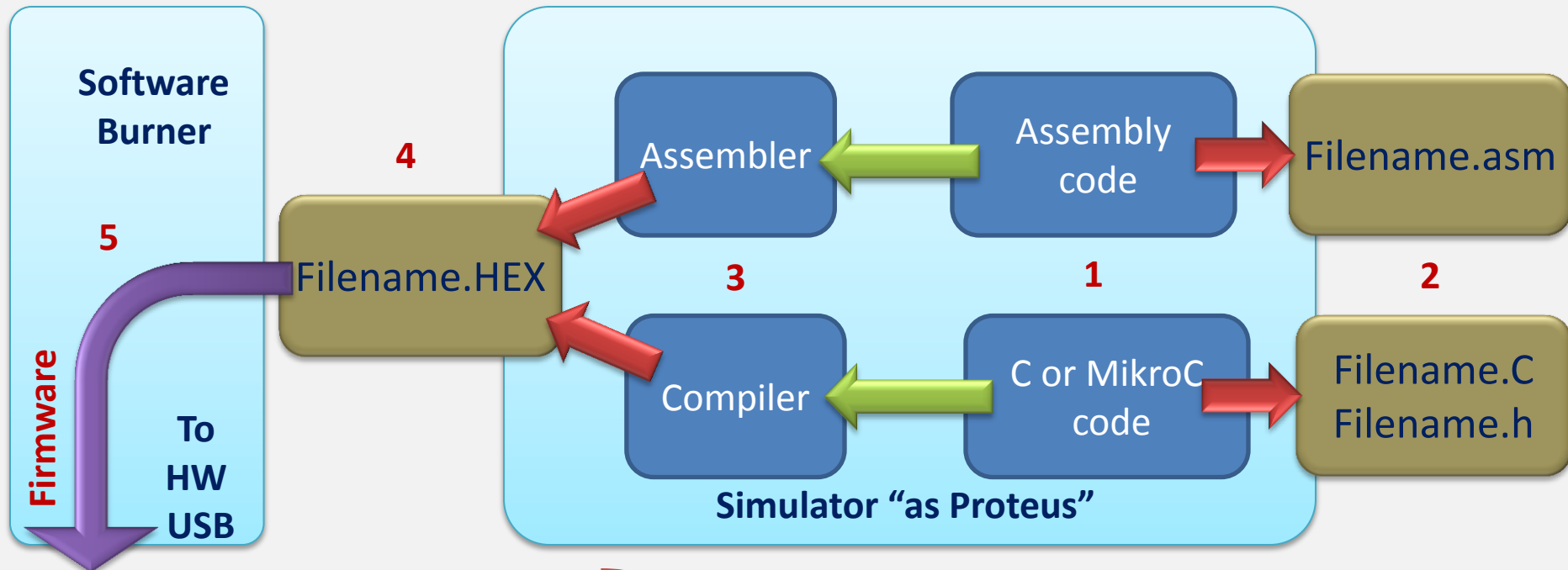
MICROPROCESSOR BASED SYSTEMS

CMP 410

LECTURE 2

<https://AssemSite8.wix.com/site8>

Microcontroller Firmware procedures



6 } **Training-KIT**

→ **To Embedded system**

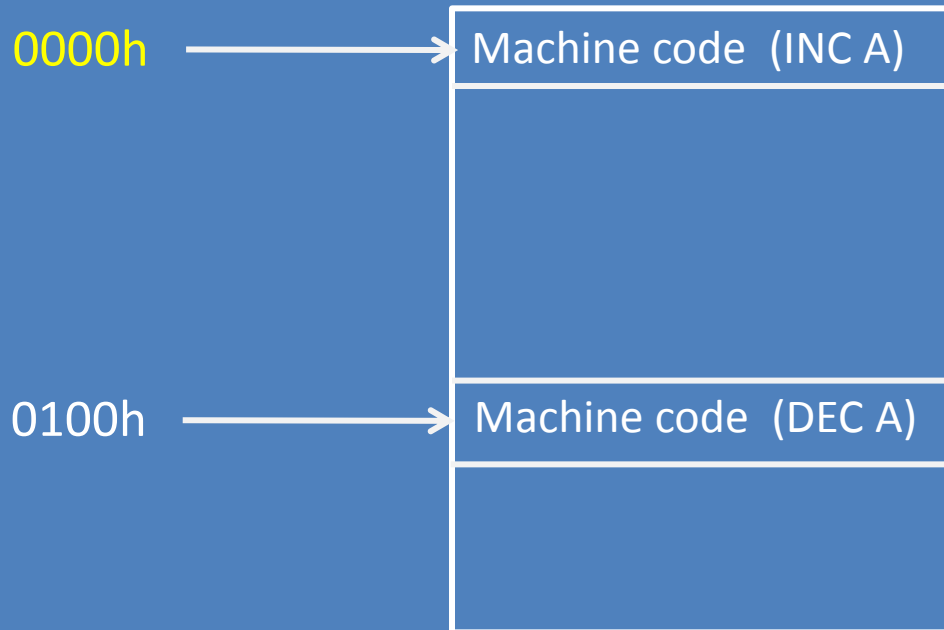
ORG

Directive used by assembler to define the start address of the current machine code at the program memory

Such as

```
ORG 0  
INC A
```

```
ORG 100h  
DEC A
```

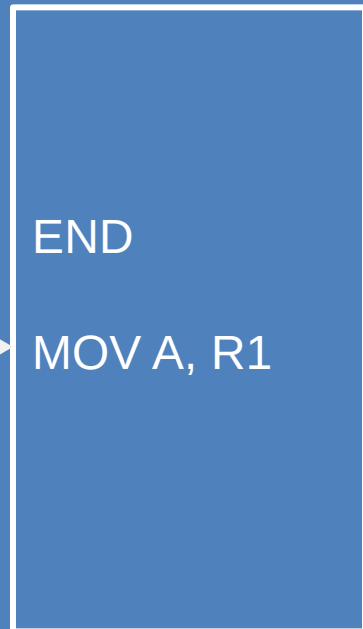


Program M

END

- Directive used by assembler to stop compiling.
- Any instruction after “End” consider as comment.

Any instruction after “End ”
Consider as comment



Program M

The machine cycle (MC)

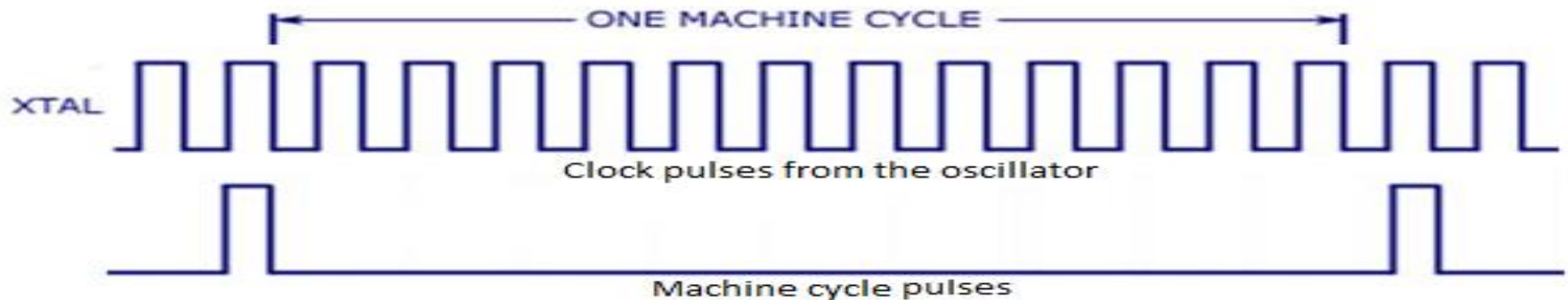
is the minimum number of CPs
(12 CPs) taken by CPU to perform
a single instruction cycle (F-D-E)

What is the difference
between **machine** code
and **machine** cycle?

The rate of machine cycles (internal clock pulses)



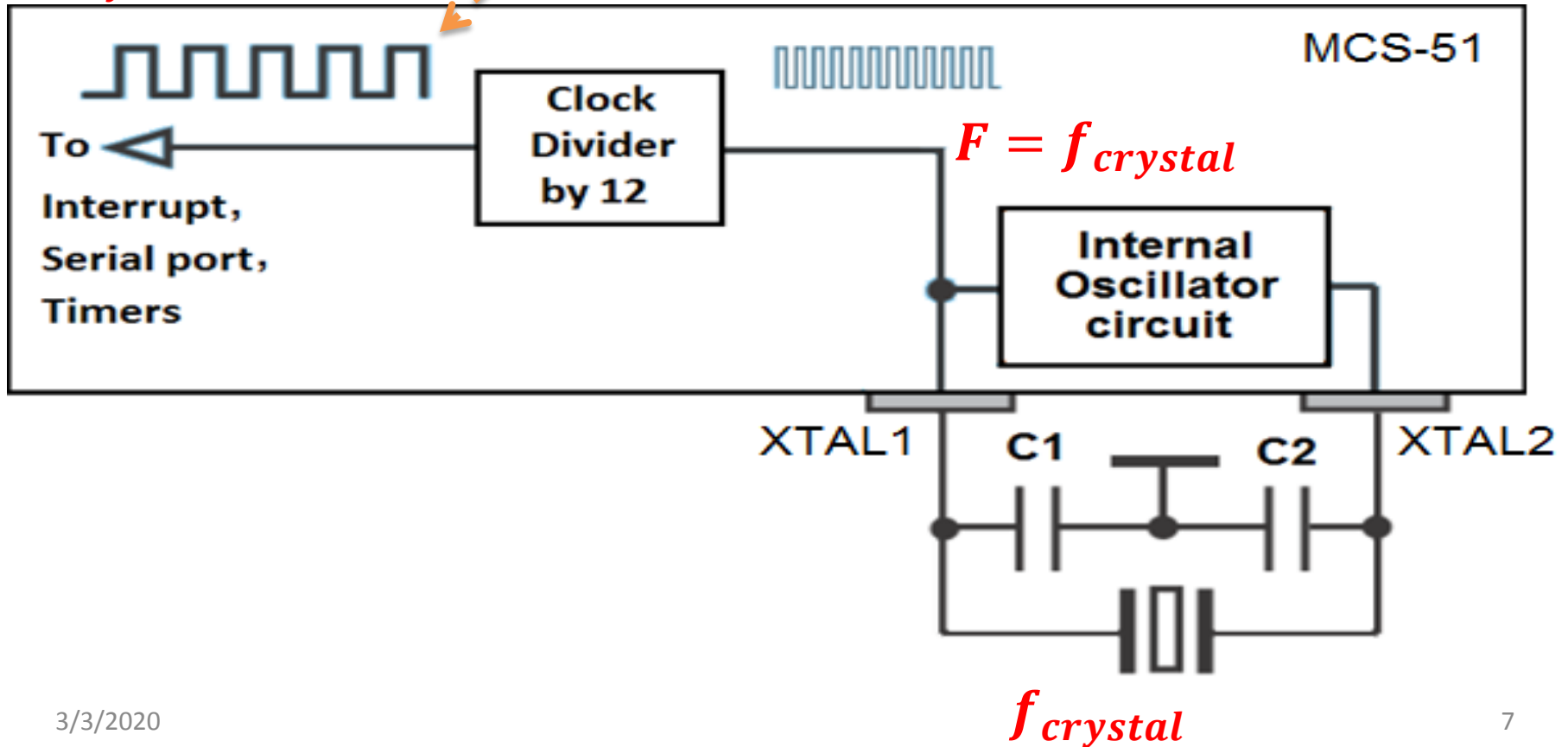
$$\text{Rate of Machine cycle} = \frac{\text{Crystal frequency}}{12} [MHz]$$



The internal clock pulse

Uses by CPU to synchronize all processes in the microcontrollers

$$f_{crystal}/12$$



EXAMPLE

Calculate the rate of the internal clock pulse of AT89C55, if the external crystal has frequency 3 MHz

The answer

$$\begin{aligned}\text{The Rate of internal clock pulse} &= \frac{\text{Crystal frequency}}{12} \\ &= \frac{3 \times 10^6}{12} = 250 \text{ KHz}\end{aligned}$$

The Execution time

Is time taken by CPU to carry out one instruction

The Run-time (Delay-time)

Is time taken by CPU to carry out program or subprogram

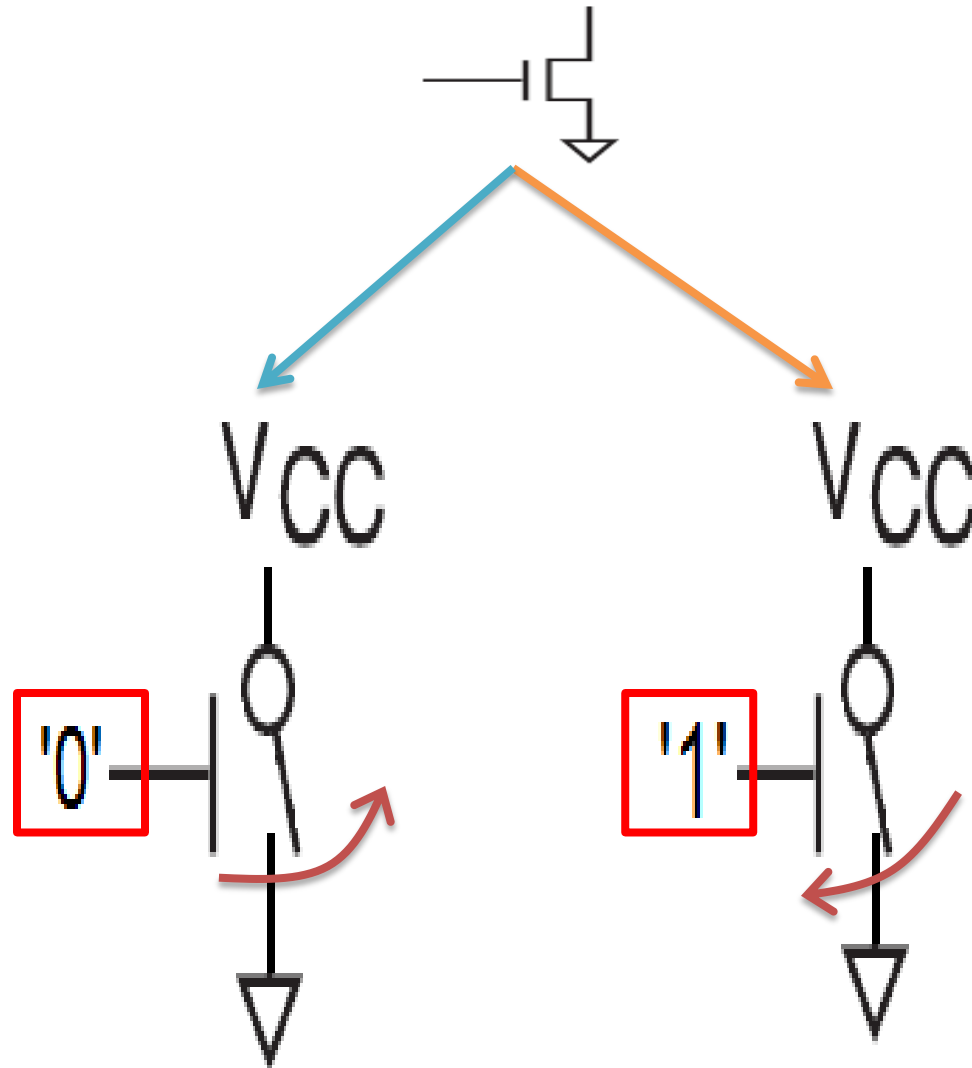
EXAMPLE

Calculate the total run-time (total delay time) for AT89C51 during execution of 3 instructions that need 1, 2 and 4 machine cycles respectively, the connected crystal has frequency equal 6 MHz.

The answer

$$\begin{aligned} \text{Delay time} &= \frac{\text{Total number of Machine cycles} \times 12}{\text{Crystal frequency}} \\ &= \frac{(4 + 2 + 1) \times 12}{6} = 14 \mu\text{sec} \end{aligned}$$

The FET as electronic switch



The internal structure of the ports

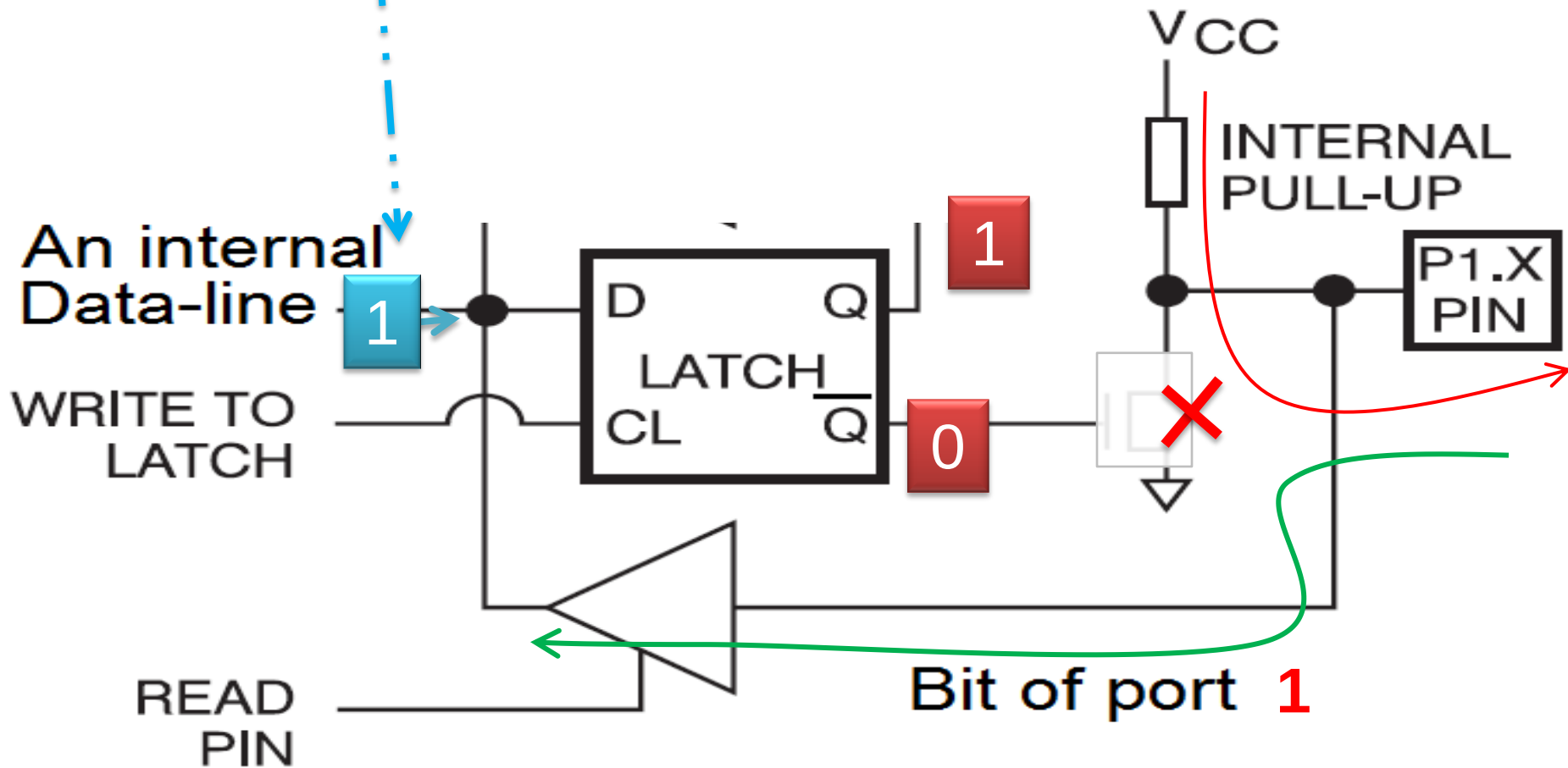
Quasi bi-directional ports



Why we write FFh to 8051 ports?

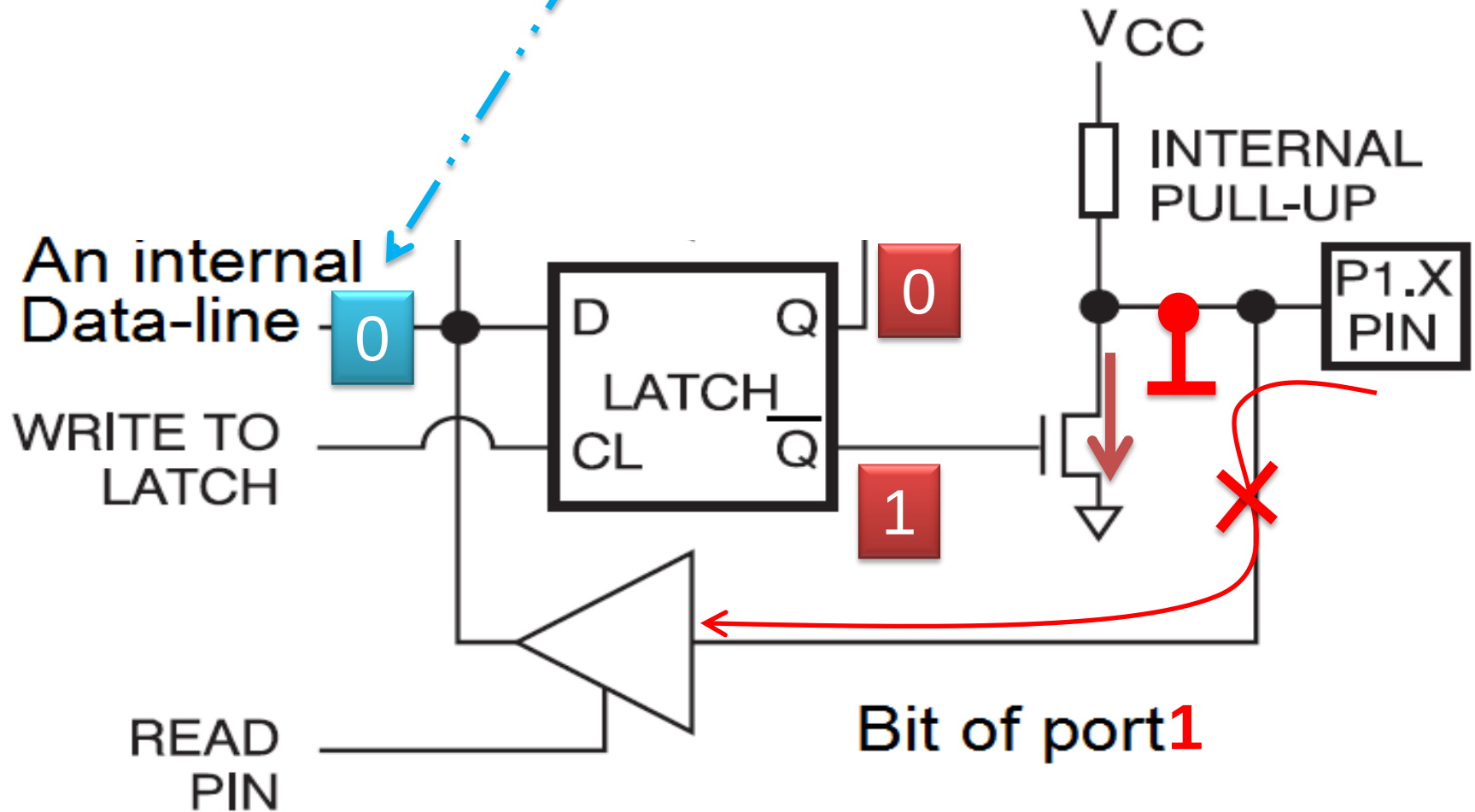
The internal structure of a pin of port1

By Setting the FF (or resetting μC)



- The bit-of-port has output “1”
- it can accept external “1” or “0” after the “**READ PIN**” is activate

By clearing the Flip-Flop



- The bit-of-port has output “0”
- it can't accept any external “1”

Why 8051's ports are Quasi bi-directional

The ports can output either '1' or '0' at any time

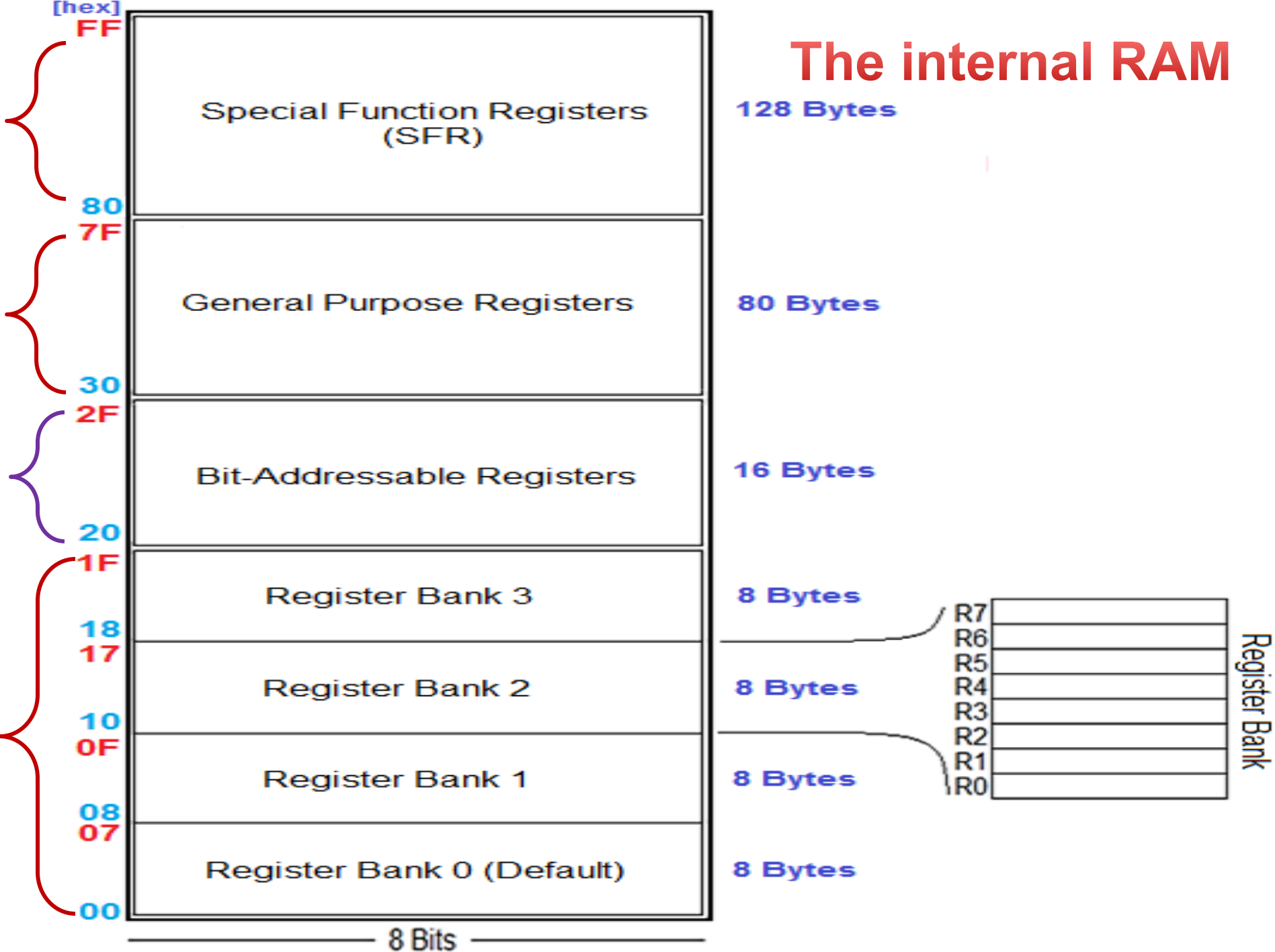
The ports can accept input '0' at any time

but

The ports can accept input '1' **only** if its flip-flop is loaded by '1'

The internal memories of 8051

The internal RAM



The special function registers

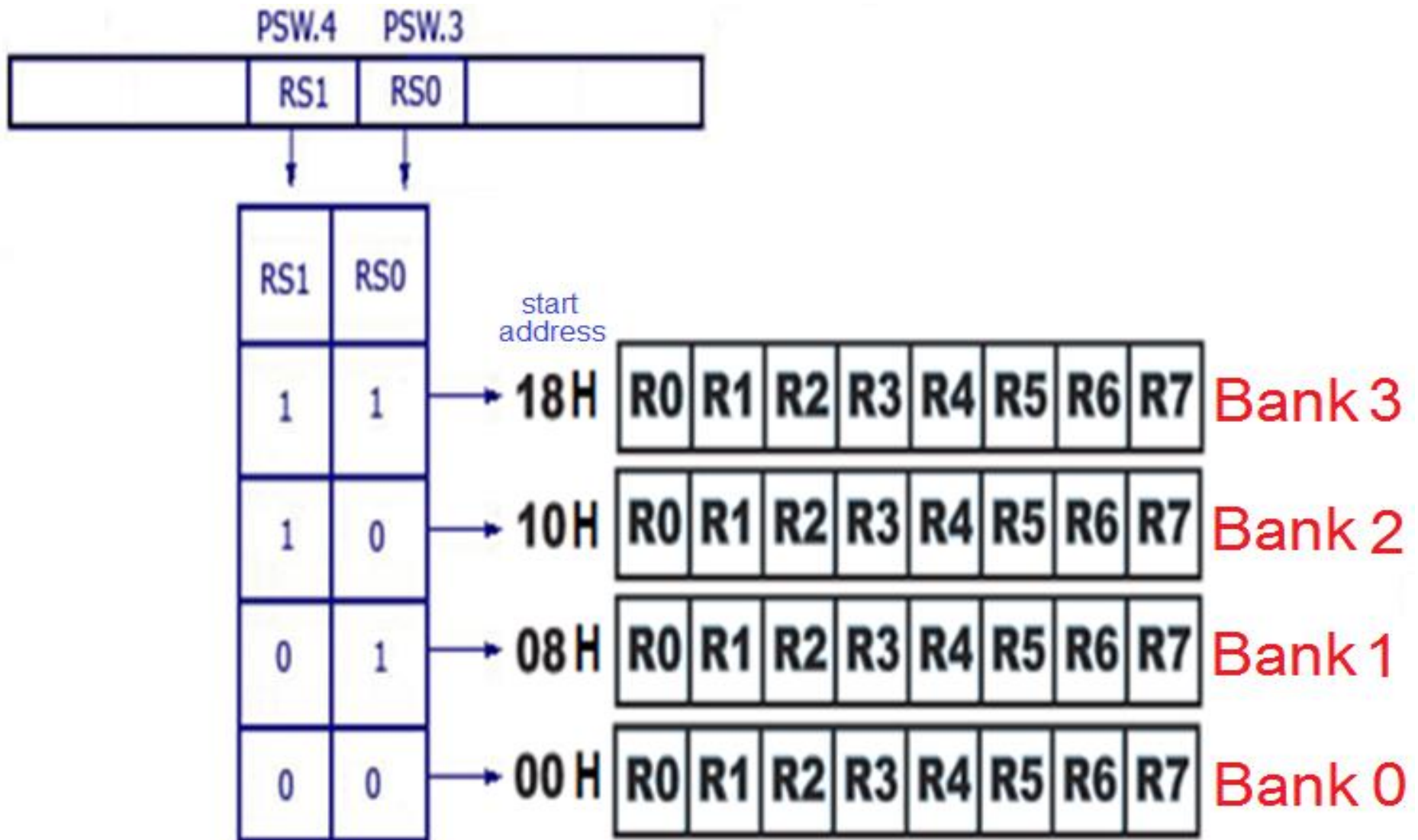
	Bit Addressable								SFR
F0	F7	F6	F5	F4	F3	F2	F1	F0	B
E0	E7	E6	E5	E4	E3	E2	E1	E0	ACC
D0	D7	D6	D5	D4	D3	D2	D1	D0	PSW
B8	-	-	-	BC	BB	BA	B9	B8	IP
B0	B7	B6	B5	B4	B3	B2	B1	B0	P3
A8	AF	-	-	AC	AB	AA	A9	A8	IE
A0	A7	A6	A5	A4	A3	A2	A1	A0	P2
99	Not Bit Addressable								SBUF
98	9F	9E	9D	9C	9B	9A	99	98	SCON
90	97	96	95	94	93	92	91	90	P1
8D	Not Bit Addressable								TH1
8C	Not Bit Addressable								TH0
8B	Not Bit Addressable								TL1
8A	Not Bit Addressable								TL0
89	Not Bit Addressable								TMOD
88	8F	8E	8D	8C	8B	8A	89	88	TCON
87	Not Bit Addressable								PCON
83	Not Bit Addressable								DPH
82	Not Bit Addressable								DPL
81	Not Bit Addressable								SP
80	87	86	85	84	83	82	81	80	P0

The Program-Status-Word Register (PSW)

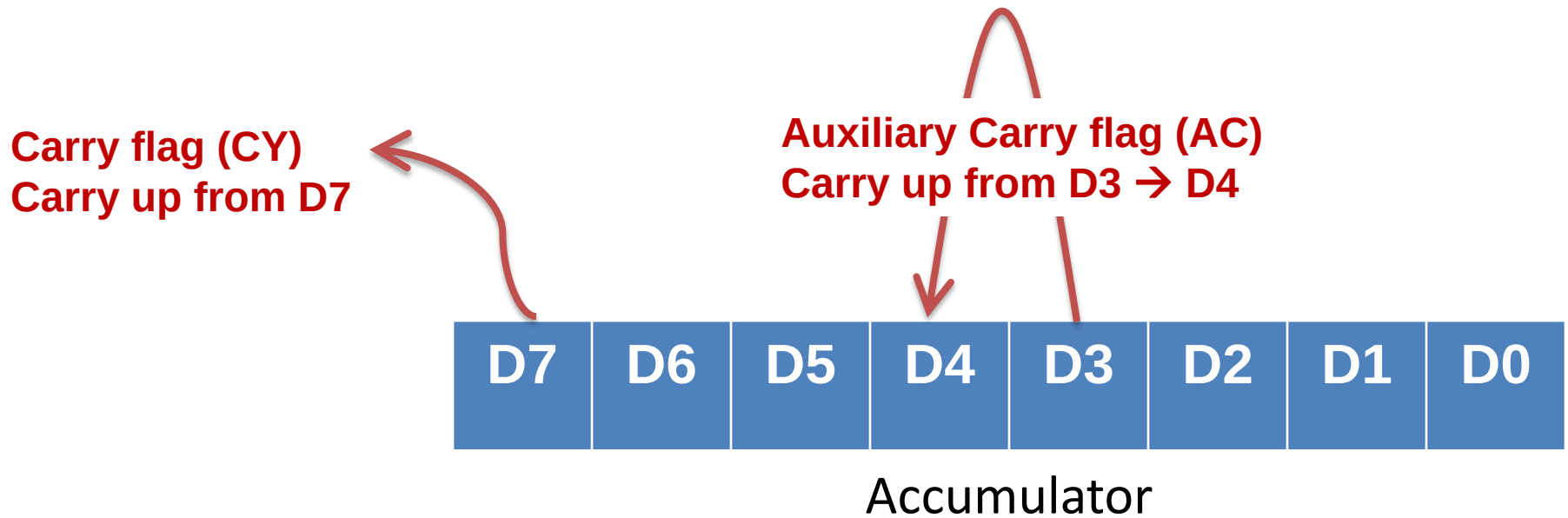


- 4 status flags (CY, AC, OV, P)
- 2 control flags (RS0, RS1)
- User bit-addressable flag (F0)
- Reversed flag (bit1)

Selection of operational bank



The Carry and Auxiliary carry flags



The parity flag (P)

$$P = \sum_{\text{bit } 0}^{\text{bit } 7} \text{ones in Accumulator}$$

Odd data → P = 1

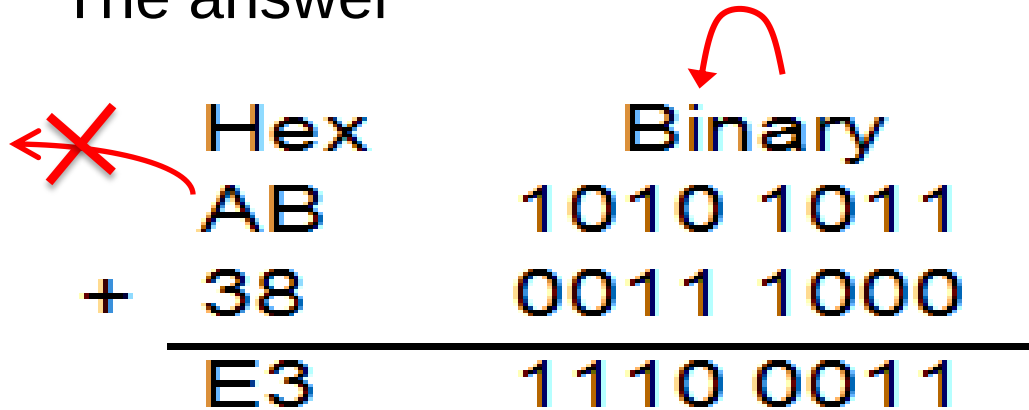
Even data → P = 0
Or
Zero data

EXAMPLE

Find the status of the flags CY, AC and P after the addition of the two digital values ABh and 38h in AT89C51.

The answer

	Hex	Binary
	AB	1010 1011
+	38	0011 1000
	E3	1110 0011



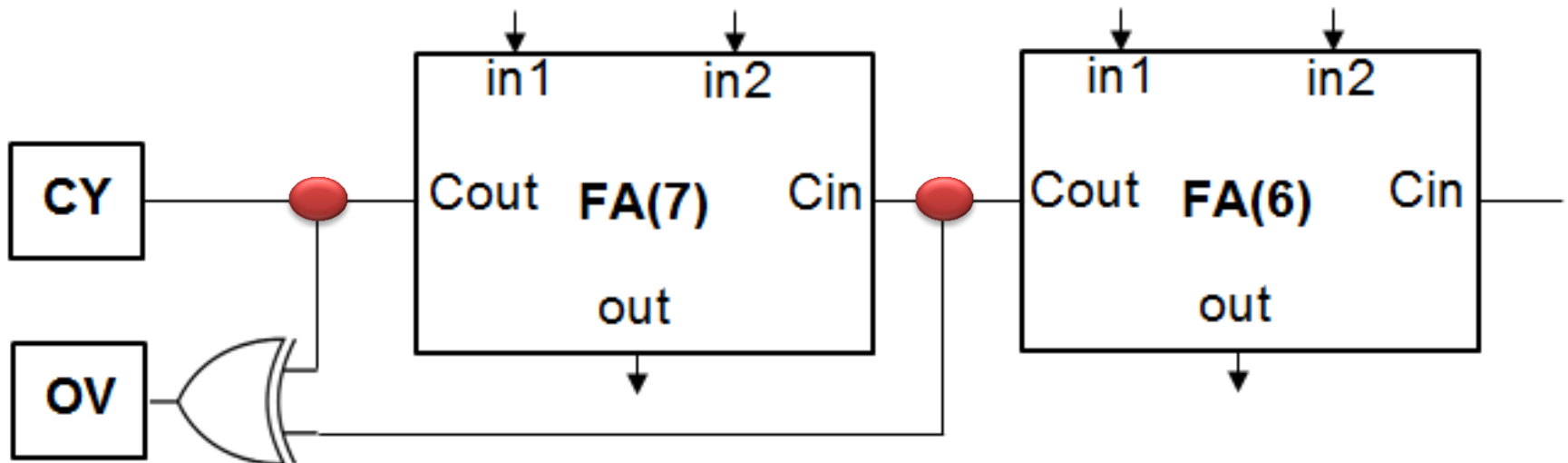
CY = 0 because there isn't carry-up from 8th bit of the final result.

AC = 1 because there is carry-up from 4th → 5th bit of the final result.

P = 1 because the Accumulator has odd number of ones (5 ones).

The Over Flow flags (OV = 1) in three cases

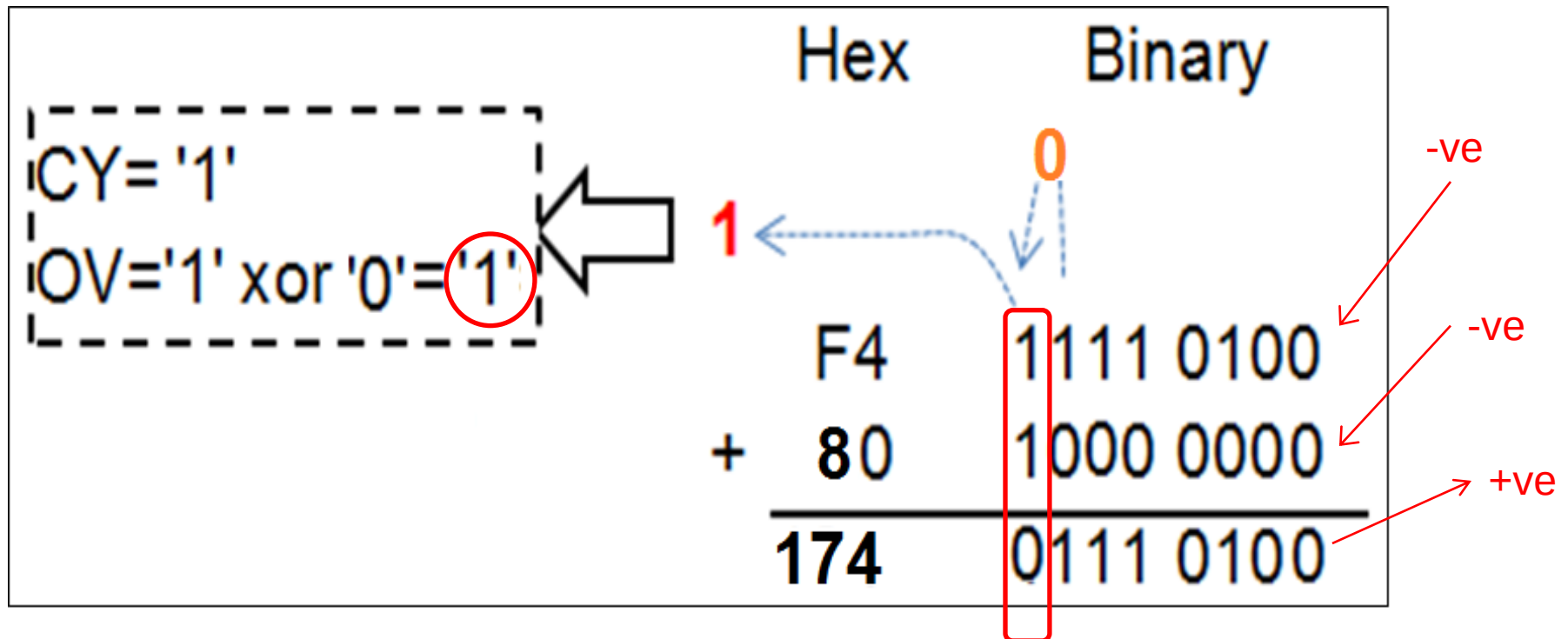
- When the divisor has value zero during the division operation.
- When the multiplication product is greater than the value "FFh".
- When an error occurs during the addition and subtraction operations for the **singed numbers** (the result outside the signed 8-bit range "-128" to "127").



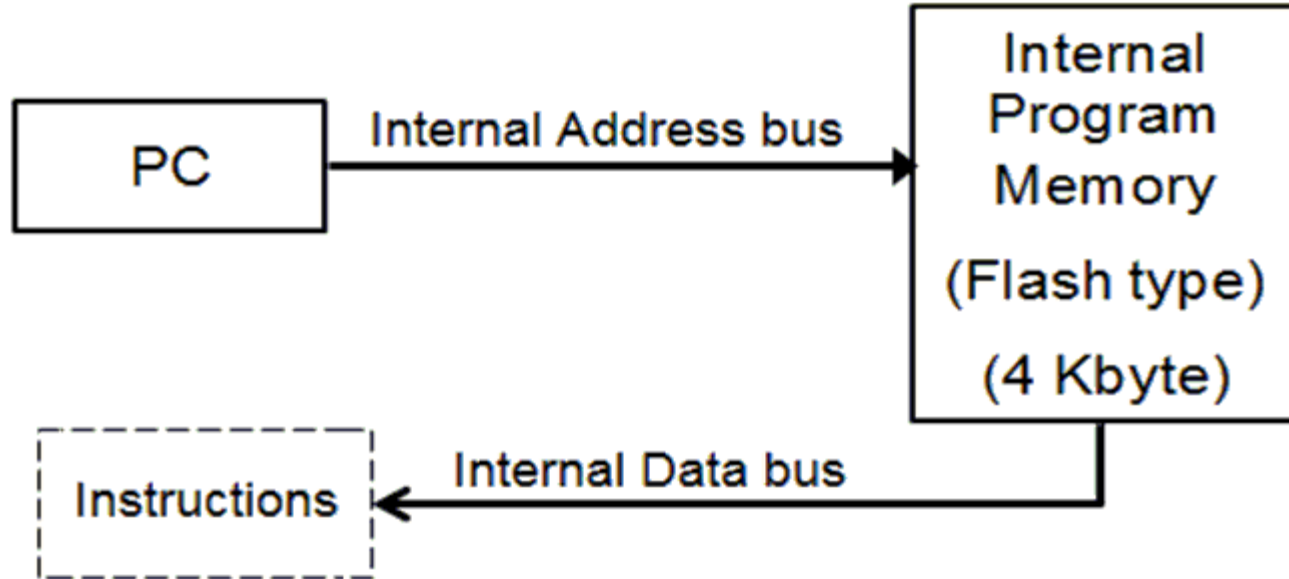
EXAMPLE

Find the status of the flags CY and OV after the addition of the two digital values F4h and 80h in the AT89C51.

The answer

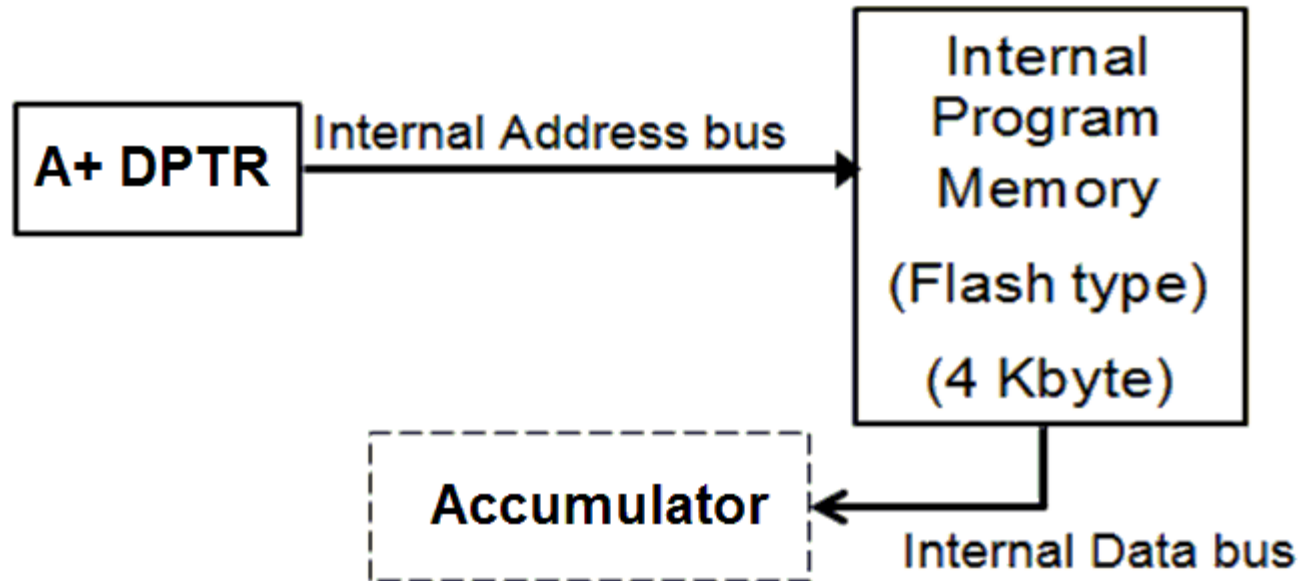


The Internal Program memory



The PC points to the memory location to fetch current instruction.

Accessing the Internal Program memory of 8051



MOVC A, @A+DPTR

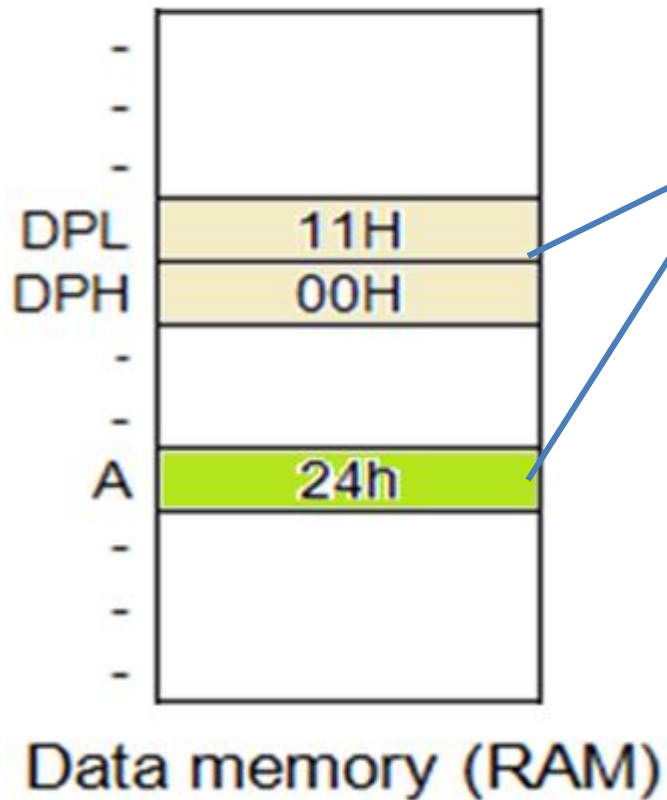
Destination

Pointers

Example

MOVC A, @A+DPTR

The pointer $\rightarrow A + DPTR = 24h + 0011h = 35h$



After executing the instruction

MOVC A, @A+DPTR

